5.2 The Characteristic Equation

– Useful info about the eigenvalues of a square matrix A is encoded in a special scalar equation called the characteristic equation of A.

Ex: Find eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.

$\mathbb{I}\,x = 0$

$\mathbb{D}$

Sol We must find all scalars λ s.t. the equation $(A - \lambda I)x = 0$ has a nontrivial solution. By the invertible matrix theorem, this problem is equivalent to finding all λ s.t. the matrix $A - \lambda I$ is not invertible, where $A - \lambda I = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$

This matrix fails to be invertible precisely when its determinant is zero. So, the eigenvalues of A are the solutions of equation:

$$\det(A - \lambda I) = \det\begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} = 0$$

$$\det(A - \lambda I) = (2-\lambda)(-6-\lambda) - 3\cdot 3 = -12 + 6\lambda - 2\lambda + \lambda^2 - 9$$

$$= \lambda^2 + 4\lambda - 21$$

$$= (\lambda - 3)(\lambda + 7)$$

If $\det(A - \lambda I) = 0$, then $\lambda = 3$ or $\lambda = -7$. So the eigenvalues are $3, -7$.

– Let A be an $n \times n$ matrix, let U be any echelon form obtained from A by row replacements and row interchanges (without scaling), let r be the number of such row interchanges. Then the determinant of A is $(-1)^r$ times the product of the diagonal entries $u_{11}, \ldots, u_{nn}$ in U. Thus,

$$\det A = \begin{cases} (-1)^r \cdot \begin{pmatrix} \text{product of} \\ \text{pivots in } U \end{pmatrix}, & \text{when } A \text{ is invertible} \\ 0, & \text{when } A \text{ is not invertible} \end{cases}$$

Ex: Compute $\det A$ for $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$

Sol

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} = U_1$$

So, $\det A = (-1)^1 (1)(-1)(-1) = -2$

A different echelon form:

$$A \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 1/3 \end{bmatrix} = U_2$$

$$\det A = (-1)^0 (1)(-6)\cdot(1/3) = -2$$

Theorem:

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then A is invertible if and only if:

s. The number 0 is *not* an eigenvalue of A.

t. The determinant of A is *not* zero.

Theorem:

Properties of Determinants

Let A and B be $n \times n$ matrices.

a. A is invertible if and only if $\det A \neq 0$.

b. $\det AB = (\det A)(\det B)$.

c. $\det A^T = \det A$.

d. If A is triangular, then $\det A$ is the product of the entries on the main diagonal of A.

e. A row replacement operation on A does not change the determinant. A row interchange changes the sign of the determinant. A row scaling also scales the determinant by the same scalar factor.

The Characteristic Equation

- A scalar λ is an eigenvalue of an $n \times n$ matrix A iff λ satisfies the characteristic equation:

$$det(A - \lambda I) = 0$$

Ex: Find the characteristic equation of $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Sol

$$\det(A - \lambda I) = \det \begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ 0 & 3-\lambda & -8 & 0 \\ 0 & 0 & 5-\lambda & 4 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix} = (5-\lambda)(3-\lambda)(5-\lambda)(1-\lambda)$$

The characteristic equation is $(5-\lambda)^2(3-\lambda)(1-\lambda) = 0$.

- It can be shown that if A is an $n \times n$ matrix, then $det(A - \lambda I)$ is a polynomial of degree n called the characteristic polynomial of A.
- In general, the (algebraic) multiplicity of an eigenvalue λ is its multiplicity as a root of the characteristic equation.

Ex: The characteristic polynomial of 6×6 matrix is $\lambda^6 - 4\lambda^5 - 12\lambda^4$. Find the eigenvalues and their multiplicities.

Sol. $\lambda^6 - 4\lambda^5 - 12\lambda^4 = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2)$

The eigenvalues are 0 (multiplicity 4), 6 (multiplicity 1) and -2 (multiplicity 1)

– In practical work, eigenvalues of any matrix larger than 2×2 should be found by a computer, unless the matrix is triangular or has other special properties. Although a 3×3 characteristic polynomial is easy to compute by hand, factoring it can be difficult.

A and B are similar.

Similarity $P^{-1}AP = B \iff PP^{-1}AP = PB \iff APP^{-1} = PBP^{-1} \iff A = PBP^{-1}$

– If A and B are $n \times n$ matrices, then A is similar to B if there is an invertible matrix P such that $P^{-1}AP = B$, or, equivalently, $A = PBP^{-1}$. Writing Q for P^{-1}, we have $Q^{-1}BQ = A$. So, B is also similar to A, and we say simply that A and B are similar. Changing A into $P^{-1}AP$ is called a similarity transformation.

Theorem:

If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Proof: If $B = P^{-1}AP$, then

$$B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(AP - \lambda P) = P^{-1}(A - \lambda I)P$$

Using the multiplicative property in previous theorem, we compute

$$\det(B - \lambda I) = \det\left[P^{-1}(A - \lambda I)P\right] = \det(P^{-1}) \cdot \det(A - \lambda I) \cdot \det(P).$$

Since $\det(P^{-1})\det(P) = \det(P^{-1}P) = \det I = 1$, $\det(B - \lambda I) = \det(A - \lambda I)$.

WARNINGS:

1. The matrices
$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$
are not similar even though they have the same eigenvalues.
2. Similarity is not the same as row equivalence. (If A is row equivalent to B, then $B = EA$ for some invertible matrix E.) Row operations on a matrix usually change its eigenvalues.